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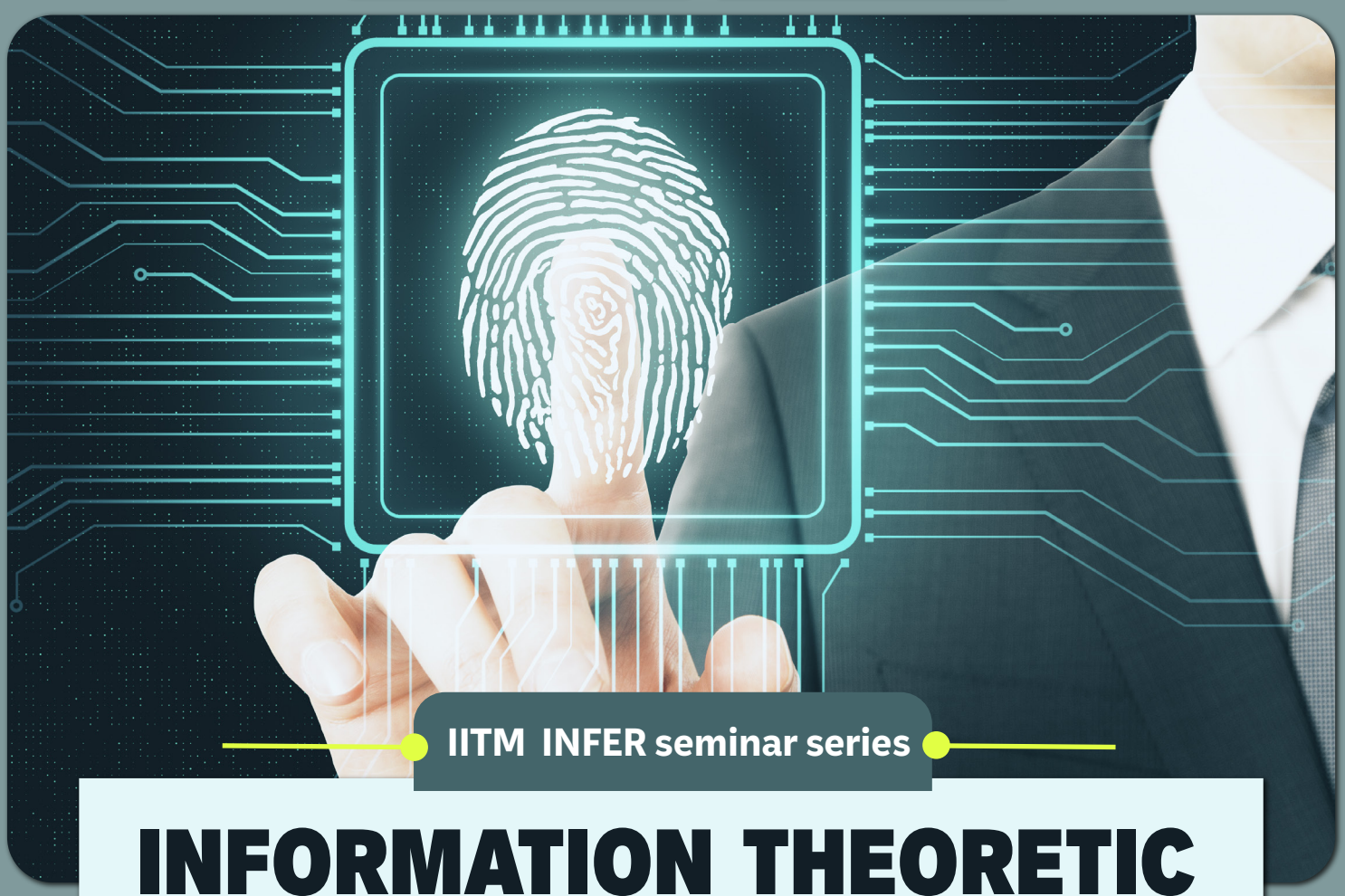

IITM INFER seminar series

INFORMATION THEORETIC DATA PRIVACY

Dept. of Electrical Engineering, IIT Madras

Session Details - April 13, 2026 at 4.00 pm | Mode - Online

YouTube Live Link: https://youtube.com/live/9txfc39SEwQ?feature=share

ABSTRACT:

Current as well as futuristic data-driven applications entail function computation or recoverability or utility, with accompanying requirements of data privacy. In such applications, a user holding data must publicly furnish attributes of the data (function recoverability or utility) to a server, while protecting the data or the data-generating mechanism (privacy). Variant formulations of function computation with privacy will be described: "predicate privacy" which requires that data or a separate and sensitive attribute of it be protected; "list privacy" which guarantees that a protected data value defies even being guessed to lie among a larger group of values; "distribution privacy" in which the workings of an underlying data-generating algorithm must remain private; and "anticollusion privacy," which guards against multiple colluding servers.
The concept of differential privacy rules the data privacy landscape. Our approach can be viewed as a complement that enables exact characterizations of optimal utility versus privacy performance tradeoffs, and specifies randomized privacy mechanisms for attaining them. This is a rich and evolving research area that raises many interesting questions.

The talk is based on joint work with **Ajaykrishnan Nageswaran** and **Rohit Premlal**.

SPEAKER

DR. PRAKASH NARAYAN

Prakash Narayan is a distinguished academic in electrical and computer engineering. He completed his B.Tech in Electrical Engineering from IIT Madras and earned his M.S. and D.Sc. from Washington University in St. Louis.

He is a Professor at the University of Maryland, College Park, with a joint appointment at the Institute for Systems Research, and is a founding member of the Maryland Hybrid Networks Center. He has held several international visiting positions at leading institutions across Europe, Israel, and India.

Dr. Narayan has played significant roles in the IEEE Information Theory community, including serving on its Board of Governors and contributing as an editor and conference organizer. He is recognized as a Fellow of the IEEE for his contributions to the field.